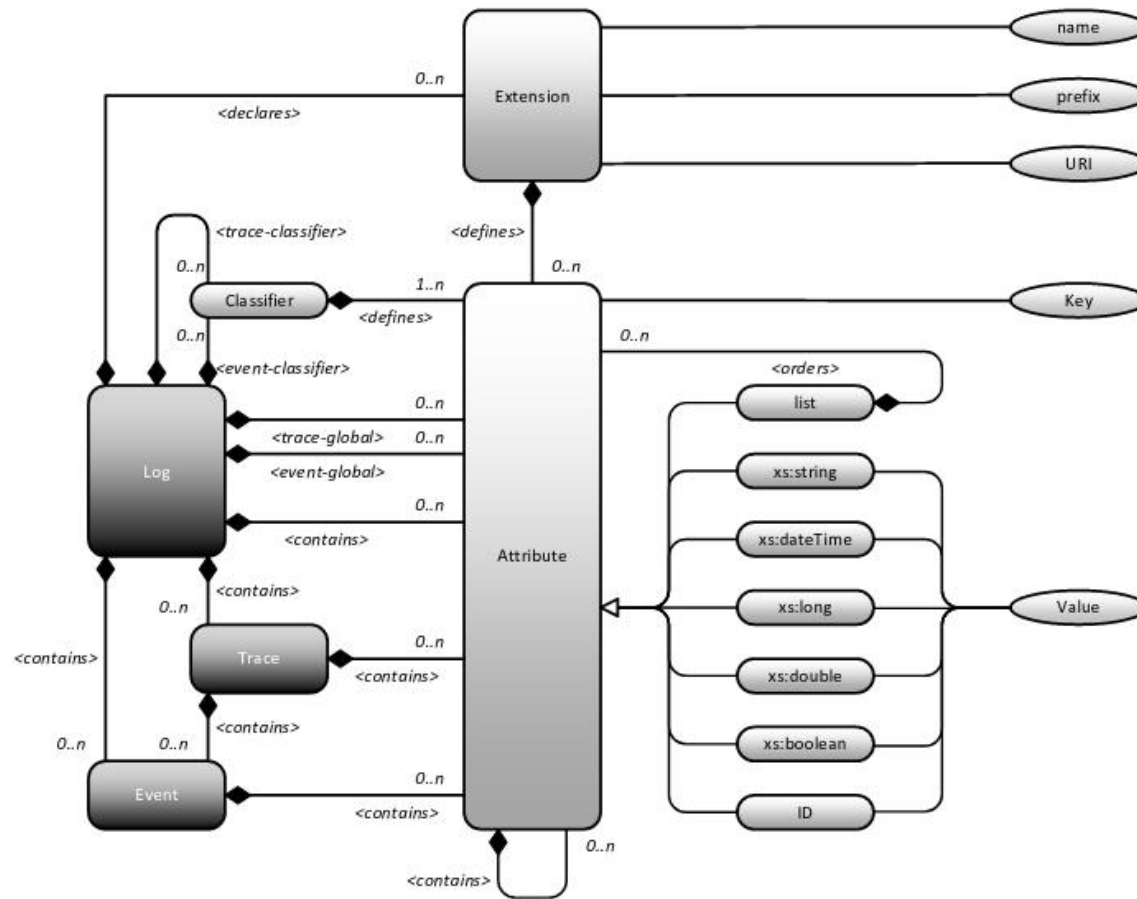


Business Process Technologies and Management

Exercises

juergen.mangler@tum.de

XES Model



HEADER

```
<log xes.version="1.0" xmlns="http://www.xes-standard.org" xes.creator="cpee.org" xes.features="nested-attributes">
  <extension name="Concept" prefix="concept" uri="http://www.xes-standard.org/concept.xesext"/>
  <extension name="Lifecycle" prefix="lifecycle" uri="http://www.xes-standard.org/lifecycle.xesext"/>
  <extension name="Identity" prefix="id" uri="http://www.xes-standard.org/identity.xesext"/>
  <extension name="Time" prefix="time" uri="http://www.xes-standard.org/time.xesext"/>
  <extension name="cpee.org" prefix="cpee" uri="http://cpee.org"/>
  <global scope="trace">
    <string key="concept:name" value="name"/>
  </global>
  <global scope="event">
    <string key="concept:name" value="name"/>
    <string key="lifecycle:transition" value="start"/>
    <date key="time:timestamp" value="1970-01-01T00:00:00.000+00:00"/>
  </global>
  <string key="lifecycle:model" value="standard"/>
  <string key="creator" value="cpee.org"/>
  <string key="library" value="cpee.org"/>
  <trace></trace>
  <trace></trace>
</log>
```

TRACE XML serialization

```
<trace>
  <string key="concept:instance" value="3"/>
  <event>
    <string key="concept:name" value="accept"/>
    <date key="time:timestamp" value="2010-12-30T14:32:00.000+01:00"/>
    <date key="id:id" value="a1"/>
    <date key="lifecycle:transition" value="start"/>
    <string key="role" value="Clerk"/>
    <string key="costs" value="50"/>
  </event>
  <event>
    <string key="concept:name" value="accept"/>
    <date key="time:timestamp" value="2010-12-30T14:35:00.000+01:00"/>
    <date key="id:id" value="a1"/>
    <date key="lifecycle:transition" value="complete"/>
    <string key="result" value="{\"decision\": true}"/>
  </event>

  ...
</trace>
```

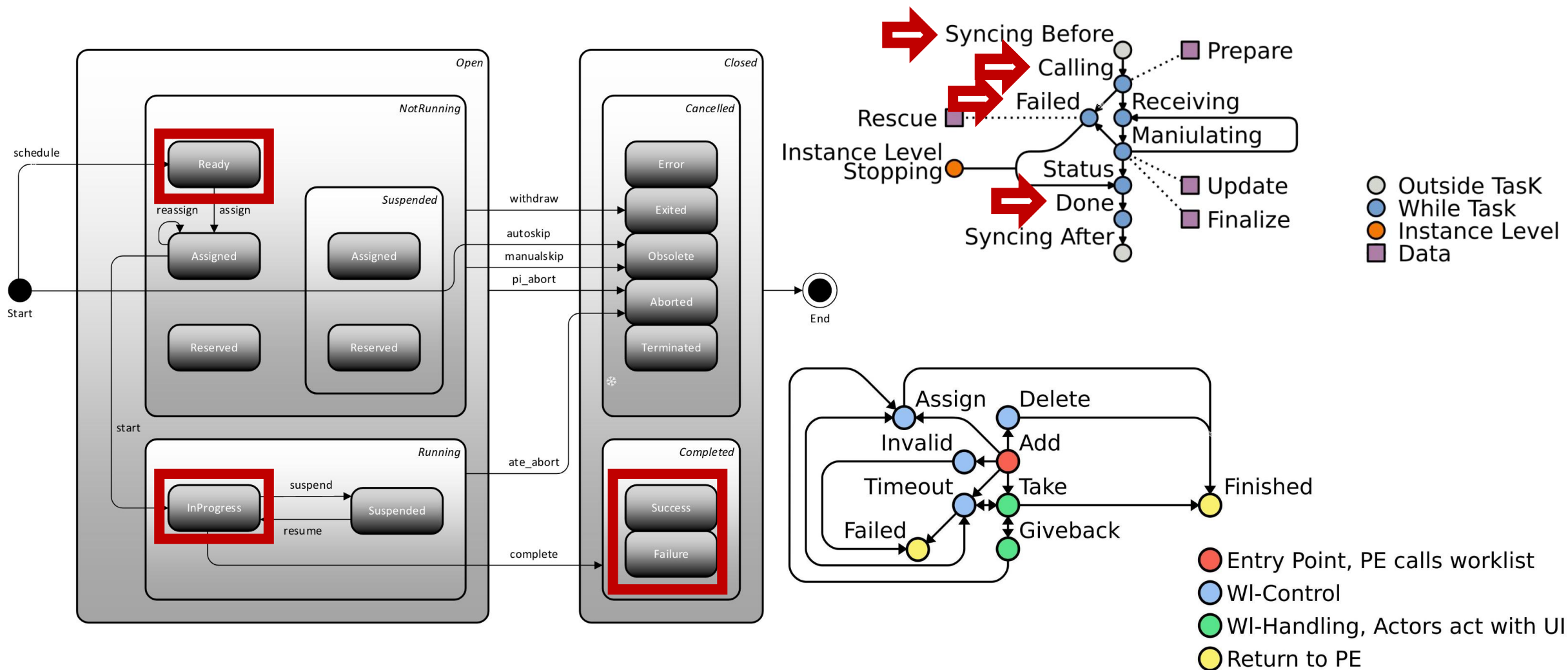
Many traces per file are possible. One trace == One instance.

TRACE YAML serialization

```
trace:
  concept:instance: 3
  event:
    concept:name: accept
    time:timestamp: '2010-12-30T14:32:00.000+01:00'
    id:id: a1
    lifecycle:transition: start
    role: Clerk
    costs: 50
  event:
    concept:name: accept
    time:timestamp: '2010-12-30T14:35:00.000+01:00'
    id:id: a1
    lifecycle:transition: complete
    result: '{"decision": true}'
  ...
```

Many traces per file are possible. One trace == One instance.

Lifecycle



Lifecycle ctd.

- In XES not much is about task, major focus on resources
- Different POV:
 - Worklist is a part of the process engine and its lifecycle
 - Worklist is just another task, each task can have its own internal lifecycle

Assignment 5 - What to do (total 20 points)?

1. Export BPMN 2.0 XML from Signavio
2. Create A Log File and Visualisation for it
 - Select appropriate **events** (not votes) from <https://cpee.org/flow/engine/1/notifications/topics/>
 - Save your testset to a file
 - Add **ONE** subscription to your testset file (edit the XML) - as a child of the the testset element. Example:

```
<subscriptions xmlns="http://riddl.org/ns/common-patterns/notifications-producer/2.0">
  <subscription xmlns="http://riddl.org/ns/common-patterns/notifications-producer/2.0" id="_student" url="https://lehre.bpm.in.tum.de/~tumuser/log.php">
    <topic id="state">
      <event>change</event>
    </topic>
    <topic id="activity">
      <event>calling</event>
      <event>failed</event>
      <event>done</event>
    </topic>
  </subscription>
</subscriptions>
```
 - All people whose matnrs end 1,2,3,4,5: the XES file should be in the «YAML» (NOT XML) serialization format but noetheless follow exactly the XES format (same fields, same structure).
 - All people whose matnrs end 6,7,8,9,0: the XES file should be in the traditional XML serialization format.
3. 10 bonus points: Semantic LOG Visualization
 - Write a second subscription that pushes the data into a custom data store (e.g. sqlite)
 - Write an independent website that shows that visualizes the semantic context of your process
 - E.g., how many burgers have been flipped, how many drinks have been served (subscribe to dataelements/change)
 - Use Diagrams, e.g. plotly
 - Show runtime information, no reload of the page necessary, the diagrams update themselves, e.g. through SSE
 - DO NOT visualize number of instances or timing of tasks, visualize data flow!

Assignment 5

Submission Artefacts:

1. Export BPMN 2.0 XML from Signavio - newest version
2. Testset with Sample Process
3. PDF with many screenshots showing the details of your solution.
4. Source Code

Deadline: as always, monday in 2 weeks, 23:59.

Please have first results in 1 week, so that we can discuss and answer questions in the course.